

C884

Statistics and Probability for Secondary Mathematics Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A class surveyed 60 families about their favorite of five fruits; the survey ran on the school's new tablet app and took two days. A student graphs the counts with touching bars and says the distribution is "skewed right." What is the best teacher response?

- A.** Explain that fruit is categorical, so a bar graph with gaps fits, and skew has no meaning because the category order is arbitrary.
- B.** Ask the student to find the mean favorite fruit to confirm the skew.
- C.** Have the class resurvey more families before graphing.
- D.** Accept the description; any graph with a long tail can be called skewed.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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2.

A district tests a homework app that costs \$14 per student license. Two hundred volunteer students are randomly assigned to use the app or not; the app group scores significantly higher. The app's vendor is based in another state. Which conclusion is justified?

- A.** The app will raise scores for every student in the district.
- B.** No conclusion is possible because the vendor is out of state.
- C.** The app likely caused the gain for students like these volunteers, but the result may not generalize to all district students.
- D.** No cause can be claimed because the students were not a random sample.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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3. [TRANSFER — reversed stem / different variable]

A researcher wants to generalize from a sample to all ninth graders in a state. Which design feature is essential?

- A.** Any sample over 1,000 students, however chosen.
- B.** Including a control group.
- C.** Randomly assigning the sampled students to treatment groups.
- D.** Selecting the sample at random from all ninth graders in the state.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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4.

A school website invites visitors to vote on whether the school day should start later, and the poll was designed by the math club with colorful graphics. Of 1,850 responses, 82% favor a later start. What is the main problem with using 82% as the view of all families?

- A. Voluntary-response bias: those with strong opinions are more likely to vote.
- B. The sample is too small to estimate a percentage.
- C. The colorful graphics may have confused voters.
- D. There is no problem; 1,850 responses guarantee an accurate result.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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5.

A counselor's table, kept in a shared spreadsheet, shows 200 students: 80 play a sport, and 60 of those passed a fitness test; of the 120 who do not play, 72 passed. What is the probability a student passed, given that the student plays a sport?

- A. 0.66
- B. About 0.45
- C. 0.75
- D. 0.30

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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6. [TRANSFER — reversed stem / different variable]

Of 200 students, 80 play a sport; 60 sport players and 72 non-players passed a fitness test. What is the probability that a student who passed plays a sport?

- A. 0.30
- B. 0.75
- C. 0.40
- D. About 0.45

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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7.

In a probability lab using a free online coin simulator, a student gets heads six times in a row with a fair coin and says tails is now “due.” The class is three weeks into the unit. What is the probability the next flip is tails?

- A. More than $1/2$, because tails is due after a streak.
- B. $1/2$
- C. $1/64$
- D. Less than $1/2$, because a free simulator must be biased.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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8. [TRANSFER — reversed stem / different variable]

Before any flips, what is the probability that a fair coin lands heads four times in a row?

- A. $1/8$
- B. $1/2$
- C. $1/4$
- D. $1/16$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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9.

A school raffle sells 500 tickets at \$5 each, printed on recycled paper by a local shop. There is one \$1,000 prize and four \$100 prizes. What is the expected net value of one ticket?

- A. \$0, because raffles are fair games.
- B. $-\$2.20$
- C. $-\$2.80$
- D. $\$2.80$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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10. [TRANSFER — reversed stem / different variable]

A game costs \$2 to play. You win a \$10 prize with probability p ; otherwise you win nothing. What value of p makes the game fair?

- A. 0.1
- B. 0.5
- C. 0.2
- D. 0.25

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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11.

Using a simulation applet recommended at a regional conference, a class finds a 95% confidence interval of 38 to 52 minutes for the mean nightly homework time of district students. Which interpretation is correct?

- A.** There is a 95% chance the sample mean is between 38 and 52.
- B.** We are 95% confident the district mean is between 38 and 52 minutes.
- C.** 95% of district students do 38 to 52 minutes of homework.
- D.** The district mean is certainly between 38 and 52 minutes.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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12.

Students test whether a spinner made in shop class lands on red more than 25% of the time. In 200 spins, red came up 62 times (31%), and they compute $p = 0.03$. The spinner was painted in school colors. At $\alpha = 0.05$, which conclusion is correct?

- A.** Reject H_0 ; there is convincing evidence the spinner lands on red more than 25% of the time.
- B.** Fail to reject H_0 , because a small p-value supports the null.
- C.** The spinner definitely favors red.
- D.** There is a 3% chance the spinner is fair.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A class surveyed 60 families about their favorite of five fruits; ~~the survey ran on the school's new tablet app and took two days.~~ A student graphs the counts with touching bars and says the distribution is "skewed right." What is the best teacher response?

Target word: correct display and language for the data type **Decides:** Favorite fruit is categorical.

A	Correct	Shape descriptions need a numeric scale on the axis.
B	Confusion set	A mean cannot be computed for categories.
C	Camouflage	Reacts to the sample detail; more data does not fix the graph type.
D	Adjacent category	Skew applies to quantitative distributions, not reorderable categories.

2. Answer C Concept 2

A district tests a homework app ~~that costs \$14 per student license.~~ Two hundred volunteer students are randomly assigned to use the app or not; the app group scores significantly higher. ~~The app's vendor is based in another state.~~ Which conclusion is justified?

Target word: scope of conclusion **Decides:** Random assignment yes; random sampling no (volunteers).

A	Absolute language	No random sample, and effects are averages, not guarantees.
B	Camouflage	Vendor location does not affect the design.
C	Correct	Assignment supports cause; volunteers limit generalizing.
D	Confusion set	Causation comes from random assignment, which was used.

3. Answer D Concept 2 · transfer

A researcher wants to generalize from a sample to all ninth graders in a state. Which design feature is essential?

Target word: what permits generalization **Decides:** Generalizing needs random selection from that population.

A	Absolute language	Size does not remove selection bias.
B	Adjacent category	A comparison feature of experiments, not of sampling.
C	Confusion set	Assignment supports cause, not generalization.
D	Correct	Random sampling supports inference to the population.

4. Answer A Concept 3

A school website invites visitors to vote on whether the school day should start later, ~~and the poll was designed by the math club with colorful graphics.~~ Of 1,850 responses, 82% favor a later start. What is the main problem with using 82% as the view of all families?

Target word: why the estimate is untrustworthy **Decides:** People chose whether to respond: voluntary response.

A	Correct	Self-selected samples overrepresent strong views.
B	Adjacent category	1,850 is large; the issue is how it was collected.
C	Camouflage	Design style is not the core sampling flaw.
D	Absolute language	A large biased sample is still biased.

5. Answer C Concept 4

A counselor's table, ~~kept in a shared spreadsheet~~, shows 200 students: 80 play a sport, and 60 of those passed a fitness test; of the 120 who do not play, 72 passed. What is the probability a student passed, given that the student plays a sport?

Target word: P(pass | sport) **Decides:** The condition limits the group to the 80 sport players.

A	Confusion set	$132 \div 200$ is the overall pass rate.
B	Inverted operation	$60 \div 132$ is P(sport pass), the reversed condition.
C	Correct	$60 \div 80$.
D	Dropped element	$60 \div 200$ is the joint probability; the condition was dropped.

6. Answer D Concept 4 • transfer

Of 200 students, 80 play a sport; 60 sport players and 72 non-players passed a fitness test. What is the probability that a student who passed plays a sport?

Target word: P(sport | pass) **Decides:** Denominator is all who passed: $60 + 72 = 132$.

A	Dropped element	$60 \div 200$ ignores the condition.
B	Inverted operation	P(pass sport): the condition is reversed.
C	Wrong stem number	$80 \div 200$ is P(sport) for everyone.
D	Correct	$60 \div 132 \approx 0.455$.

7. Answer B Concept 5

In a probability lab ~~using a free online coin simulator~~, a student gets heads six times in a row with a fair coin and says tails is now "due." ~~The class is three weeks into the unit.~~ What is the probability the next flip is tails?

Target word: next-trial probability **Decides:** Fair coin flips are independent.

A	Confusion set	The gambler's fallacy.
B	Correct	Past flips do not change a fair coin.
C	Adjacent category	$(1/2)^6$ is the chance of the streak itself, not the next flip.
D	Camouflage	The stem states a fair coin; the tool detail does not decide it.

8. Answer D Concept 5 • transfer

Before any flips, what is the probability that a fair coin lands heads four times in a row?

Target word: probability of a streak **Decides:** Independent trials multiply: $(1/2)^4$.

A	Dropped element	$(1/2)^3$: one factor dropped.
B	Confusion set	Single-flip probability, not the sequence.
C	Inverted operation	$1 \div 4$ flips instead of multiplying the chances.
D	Correct	$(1/2)^4 = 1/16$.

9. Answer B Concept 6

A school raffle sells 500 tickets at \$5 each, ~~printed on recycled paper by a local shop~~. There is one \$1,000 prize and four \$100 prizes. What is the expected net value of one ticket?

Target word: expected net value **Decides:** Expected winnings $(1,000 + 400)/500 = \$2.80$, minus \$5 cost.

A	Absolute language	Fundraisers are designed to have negative expected value for buyers.
B	Correct	$\$2.80 - \5.00 .
C	Wrong stem number	Counts only one \$100 prize: $(1,100/500) - 5$.
D	Dropped element	Expected winnings only; the ticket cost is dropped.

10. Answer C Concept 6 · transfer

A game costs \$2 to play. You win a \$10 prize with probability p ; otherwise you win nothing. What value of p makes the game fair?

Target word: solve for probability **Decides:** $10p - 2 = 0$.

A	Wrong stem number	1/10 uses only the prize amount.
B	Confusion set	Treats “fair” as a 50–50 chance.
C	Correct	$10p = 2$, so $p = 0.2$.
D	Dropped element	Sets $8p = 2$ with the \$8 net prize, dropping the $(1 - p)$ factor on the \$2 loss.

11. Answer B Concept 7

Using a simulation applet ~~recommended at a regional conference~~, a class finds a 95% confidence interval of 38 to 52 minutes for the mean nightly homework time of district students. Which interpretation is correct?

Target word: meaning of the interval **Decides:** The interval estimates the population mean.

A	Adjacent category	The sample mean is the center of the interval; it is always inside.
B	Correct	The method captures the true mean in about 95% of samples.
C	Confusion set	Describes individuals, not the mean.
D	Absolute language	Confidence is not certainty.

12. Answer A Concept 8

Students test whether a spinner ~~made in shop class~~ lands on red more than 25% of the time. In 200 spins, red came up 62 times (31%), and they compute $p = 0.03$. ~~The spinner was painted in school colors.~~ At $\alpha = 0.05$, which conclusion is correct?

Target word: decision and wording **Decides:** $p = 0.03 \leq 0.05$.

A	Correct	$p \leq \alpha$ leads to rejecting H_0 .
B	Inverted operation	The decision rule is reversed.
C	Absolute language	A test gives evidence, not proof.
D	Confusion set	p assumes H_0 is true; it is not the probability H_0 is true.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Bar graph vs. histogram	Categorical data go in bar graphs (order arbitrary, gaps between bars); quantitative data go in histograms, where shape words like skew have meaning.	Describing the skew or mean of categorical data.	Data that are names or categories rather than measured numbers.
2	Random sampling vs. random assignment	Random sampling lets you generalize to the population; random assignment lets you claim cause and effect.	Swapping the two, or thinking one gives both.	Volunteers randomly split into groups, or a random sample with no treatment.
3	Sampling bias	Voluntary-response and convenience samples are biased; a larger biased sample is still biased.	Believing a big sample size fixes a bad collection method.	People chose to respond, or were the easiest to reach.
4	Conditional probability from a two-way table	$P(A B) = \text{count of A and B} \div \text{count of B}$; the condition sets the denominator.	Dividing by the grand total or by the wrong row or column.	“Of those who...” or “given that...”
5	Independence and the gambler’s fallacy	For independent trials, past outcomes do not change the next probability; a streak’s probability is the product of trial probabilities.	Thinking an outcome is “due” after a streak.	A run of the same result followed by “next”.
6	Expected value	Expected value = $\Sigma(\text{value} \times \text{probability})$; a fair game has expected net value 0.	Averaging outcomes without weighting, or forgetting the cost to play.	Prizes, costs, and chances in one stem.
7	Interpreting a confidence interval	A 95% CI gives plausible values for a population parameter; the method captures the true value in about 95% of samples.	Reading it as the range of 95% of individual values or as certainty.	“Which interpretation of the interval is correct?”
8	p-values and decisions	The p-value is the chance of results at least this extreme if H_0 is true; if $p \leq \alpha$, reject H_0 .	Reading p as the probability H_0 is true, or reversing the decision rule.	A p-value and an α in the same stem.

Course scope (public description): Statistics and Probability for Secondary Mathematics Teaching explores important conceptual underpinnings, common misconceptions and students’ ways of thinking, appropriate use of technology, and instructional practices to support and assess the learning of statistics and probability. Secondary teachers should have a deep understanding of summarizing and representing data, study design and sampling, probability, testing claims and drawing conclusions, and the historical development of content and perspectives from diverse cultures. Calculus I is a prerequisite for this course.